NANYANG TECHNOLOGICAL UNIVERSITY

School of Social Sciences

HE3002 Macroeconomics III

LECTURE 6
The IS-MPR-PC Model I

Lec 6-8 Quiz 3

8 Nov

LT IA

AD AS

INTRODUCTION

MR: Flexible prices $\pi^e \neq 0$

1. We are going to revisit some of the ideas from IS-LM and AD-AS models and expand on them in a number of ways:

   $Ms = Md \Rightarrow i$

   (a) Rather than the traditional LM curve, we will describe monetary policy in a way that is more consistent with how it is now implemented, i.e. we will assume the central bank follows a rule MPR that dictates how it sets nominal interest rates. We will focus on how the properties of the monetary policy rule influence the behaviour of the economy. $CB \rightarrow i$ Investment (firms) $\rightarrow r$ Note: $i = r + \pi^e$

   (b) We will have a more careful treatment of the roles played by real interest rates. SR: $\pi^e = 0$ P is fixed

   (c) We will focus more on the role of the public's inflation expectations. π^e

   (d) We will model the zero lower bound on interest rates and discuss its implications for policy. ZLB GFC 2008 $i = 0$

2. Our model is going to have three elements to it: Real variables

   AS (a) A **Phillips curve (PC)** describing how inflation (π) depends on output (y).

   AD { (b) An **IS curve (IS)** describing how output (y) depends upon interest rates (r).

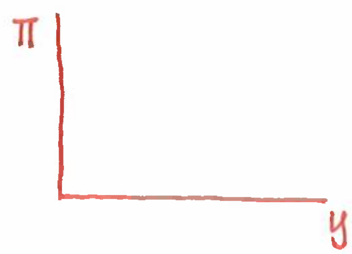
   (c) A **monetary policy rule (MPR)** describing how the central bank sets interest rates (i) depending on inflation (π) and/or output (y).

3. Putting these three elements together, we will call it the IS-MPR-PC model (i.e. The Income-Spending/Monetary Policy Rule/Phillips Curve model).

   2 Equations: IS-MPR & AS Unknowns: π & y

4. We will describe the model with equations and will also merge together the second two elements (the IS curve and the monetary policy rule) to give a new IS-MPR curve that can be combined with the Phillips curve to use graphs to illustrate the model's properties.

Endo variables: π & y



THE EVIDENCE OF PHILLIPS CURVE

1. In macroeconomics, there are important trade-offs facing governments when they implement policy. One of these relates to a trade-off between desired outcomes for inflation and output. y π
2. What form does this relationship take?
3. Back when macroeconomics was a relatively young discipline, in 1958 a study by the LSE's A.W. Phillips seemed to provide the answer.
4. Phillips documented a strong negative relationship between wage inflation (πw) and unemployment (u): Low unemployment was associated with high inflation, presumably because tight labour markets stimulated wage inflation.

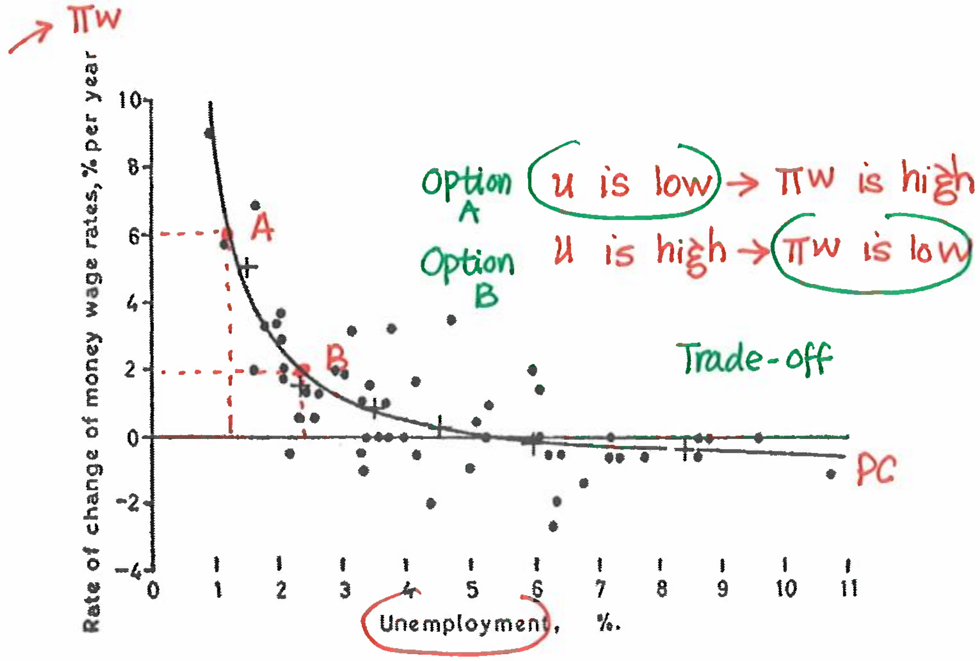


Figure 1: A. W. Phillips curve UK

5. Figure 1 shows one of the graphs from Phillips's paper illustrating the kind of relationship he found.
6. In 1960, a paper by MIT economists Robert Solow and Paul Samuelson replicated these findings for the US and emphasised that the relationship also worked for price inflation. π
7. Figure 2 reproduces the graph in their paper describing the relationship they found.
8. The Phillips curve quickly became the basis for the discussion of macroeconomic policy decisions.

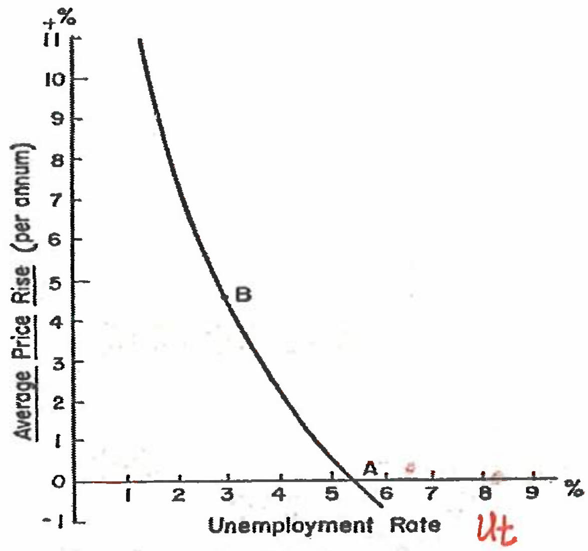


FIGURE 2

MODIFIED PHILLIPS CURVE FOR U.S.

This shows the menu of choice between different degrees of unemployment and price stability, as roughly estimated from last twenty-five years of American data.

Figure 2: Solow and Samuelson's description of the Phillips curve

9. Economists advised that governments faced a trade-off: Lower unemployment could be achieved, but only at the cost of higher inflation. ASSA

Monetarist AEA Jan

10. However, Milton Friedman's 1968 presidential address to the American Economic Association produced a well-timed and influential critique of the thinking underlying the Phillips curve. W - Nominal wage $(\frac{W}{P})$ - Real wage

11. Friedman pointed out that it was expected real wages that affected wage bargaining. If low unemployment means workers have a strong bargaining position, then high nominal wage inflation on its own is not good enough: They want nominal wage inflation greater than price inflation. Workers $L^s = L^s(\frac{W}{P^e})$ Firms $L^d = L^d(\frac{W}{P})$

expected price Actual price

12. Friedman argued that if policy-makers tried to exploit an apparent Phillips curve trade-off, then the public would get used to high inflation and come to expect it. Inflation $\pi^e \uparrow$ expectations would move up and the previously-existing trade-off between inflation and output would disappear. $\pi^e \uparrow \rightarrow$ PC shifts up $\rightarrow \pi$ & u disappears

13. In particular, he put forward the idea that there was a **natural rate of unemployment** u_n and that attempts to keep unemployment below this level could not work in the long run.

$u < u_n \rightarrow \pi \uparrow$ $u < u_n \Rightarrow y > y_n$ Expansionary gap $\pi \uparrow$

14. Monetary and fiscal policies in the 1960s were very expansionary around the world, partly because governments following Phillips curve recipes chose booming economies with low unemployment at the expense of somewhat higher inflation. option A Inflationary $\rightarrow \pi^e \uparrow$

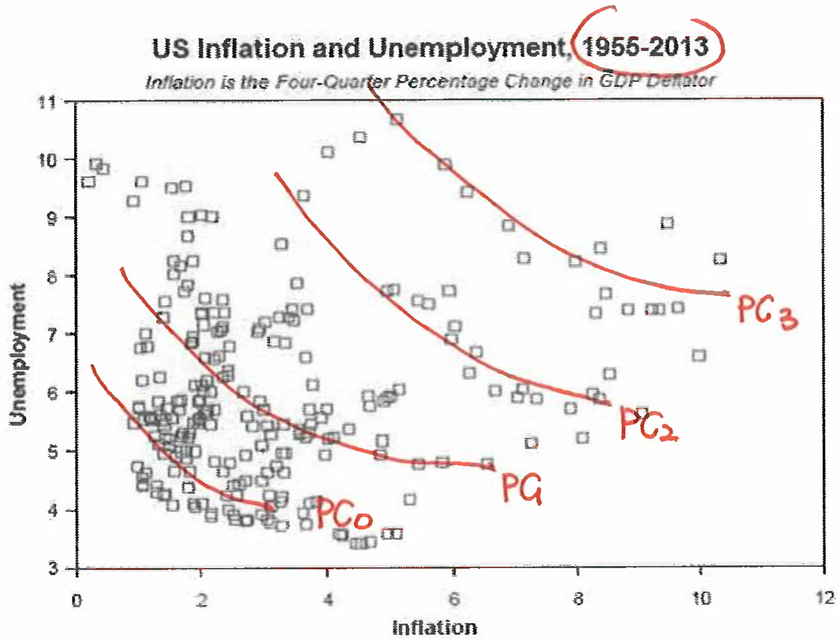


Figure 3: The failure of the Phillips curve

15. At first, the Phillips curve seemed to work: Inflation rose and unemployment fell.

16. However, as the public got used to high inflation, the Phillips trade-off got worse.

17. By the late 1960s inflation was rising even though unemployment had moved up.

18. This stagflation combination of high inflation and high unemployment got even worse in the 1970s. $\pi\uparrow$ & $u\uparrow$

19. This was exactly what Friedman had predicted.

20. Today, the data no longer show any sign of a negative relationship between inflation and unemployment. If fact, the correlation is positive: The original formulation of the Phillips curve is widely agreed to be wrong.

THE PHILLIPS CURVE

1. Our first element is an expectations-augmented (1) Phillips curve which we will formulate as a relationship in which inflation (π_t) depends on inflation expectations (π^e), the gap between output and its **natural** level and a temporary inflationary shock.

$y_t - y_t^*$ (2) $\qquad \varepsilon^{\pi}$ $\qquad y_t^*$ – output at potential.

2. Our equation for this is the following:

ε – shock

$$\pi_t = \pi_t^e + \gamma(y_t - y_t^*) + \epsilon_t^\pi \tag{1}$$

ε_t^{π} – inflationary shock

where π_t represents inflation at time t.

$\gamma > 0$ sensitivity of π_t to change in $(y_t - y_t^*)$

$1 \cdot \pi_t^e$

$\pi_t = \pi_t^e + \gamma(y_t - y_t^*) + \varepsilon_t^{\pi}$ % Δ in W = % Δ in pe

3. The equation states that inflation at time t depends on three factors:

$\pi_t^e \uparrow$ by 1 point $\rightarrow \pi_t \uparrow$ by 1 pt

(a) **Inflation expectations,** π_t^e: A 1 point increase in inflation expectations raises inflation by exactly 1 point. This is because we are assuming that people bargain over real wages and higher expected inflation translates one-for-one into their wage bargaining, which in turn is passed into price inflation. $\pi_t^e \uparrow$ by 1% $\rightarrow (\frac{W}{pe}) \downarrow \rightarrow$

y_t y_t^* or y_n

(b) **Output gap,** $(y_t - y_t^*)$: This is the gap between GDP at time t, as represented by y_t, and what we will term the natural level of output, which we term y_t^*. This is the level of output at time t that would be consistent with unemployment equalling its natural rate. We would expect this natural level of output to gradually increase over time as productivity levels improve. The coefficient γ (gamma) describes exactly how much inflation is generated by a 1 percent increase in the gap between output and its natural rate.

$y_t^* \uparrow$ over time

oil price

(c) **Temporary inflationary shocks** ϵ_t^{π}. This captures supply shocks like a temporary increase in the price of imported oil can drive up inflation for a while. To capture these kinds of temporary factors, we include an inflationary shock term.

$\pi_t^e \uparrow$ by 1% $\rightarrow (\frac{W}{pe}) \downarrow \rightarrow$ to maintain $(\overline{\frac{W}{pe}}) \rightarrow W \uparrow$ by 1% $\rightarrow$

cost of production $\uparrow$ by 1% $\rightarrow$ if $\bar{P}$, profit margin $\downarrow$ by 1% $\rightarrow$

to maintain profit margin, $P \uparrow$ by 1%.

$\Delta \pi_t^e = \Delta \pi_t$

$y_t - y_t^* \leftarrow$ output gap when $(y_t - y_t^*) > 0 \rightarrow y_t > y_t^*$

$\rightarrow$ Expansionary gap

$\rightarrow \pi_t \uparrow$

$(y_t - y_t^*) < 0 \rightarrow y_t < y_t^*$

$\rightarrow$ Recessionary gap

$\rightarrow \pi_t \downarrow$

4. Figure 4 shows how to describe our Phillips curve equation in a graph.

5. The graph shows a positive relationship between inflation and output. The key points to notice are the markings on the two axes indicating what happens when output is at its natural rate.

6. This graph illustrates the case when there are no temporary inflationary shocks so ϵ_t^π. In this case, the Phillips curve is just:

$$\pi_t = \pi_t^e + \gamma(y_t - y_t^*) \tag{2}$$

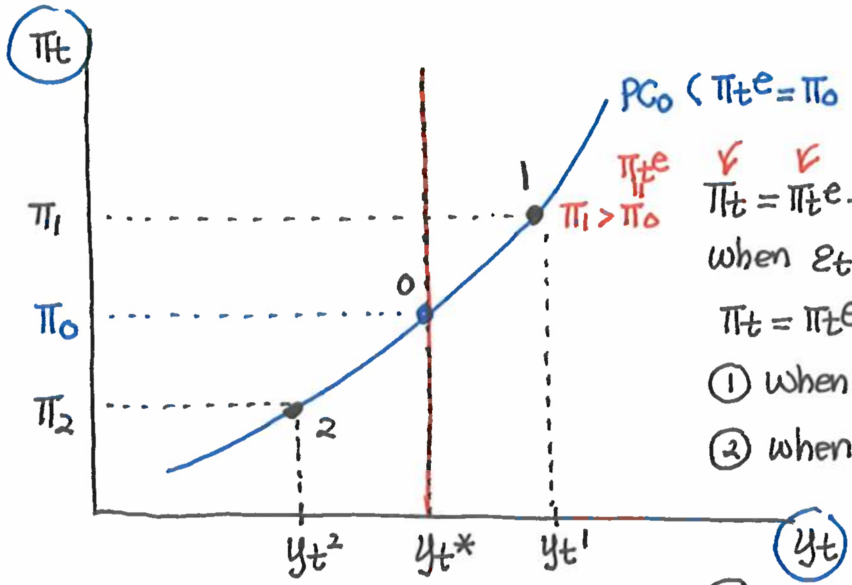
7. So when $y_t = y_t^*$ we get $\pi_t = \pi_t^e$. This is a key aspect of this graph.

8. The curve can move up or down depending on what happens to the inflationary shocks, ϵ_t^π, and with inflation expectations.

[**Figure 4:** The Phillips curve with $\epsilon_t^\pi = 0$]

Slope of PC $= \left.\frac{d\pi_t}{dy_t}\right|_{PC} = \gamma$

$\gamma > 0$



PC_0 ($\pi_t^e = \pi_0$, $\varepsilon_t^\pi = 0$)

$\pi_1 > \pi_0$

$\pi_t = \pi_t^e + \gamma(y_t - y_t^*) + \varepsilon_t^\pi$

when $\varepsilon_t^\pi = 0$,

$\pi_t = \pi_t^e + \gamma(y_t - y_t^*)$

① when $y_t = y_t^*$, $\pi_t = \pi_t^e$

② when $y_t > y_t^*$, $\pi_t > \pi_t^e$

y_t (Not u_t)

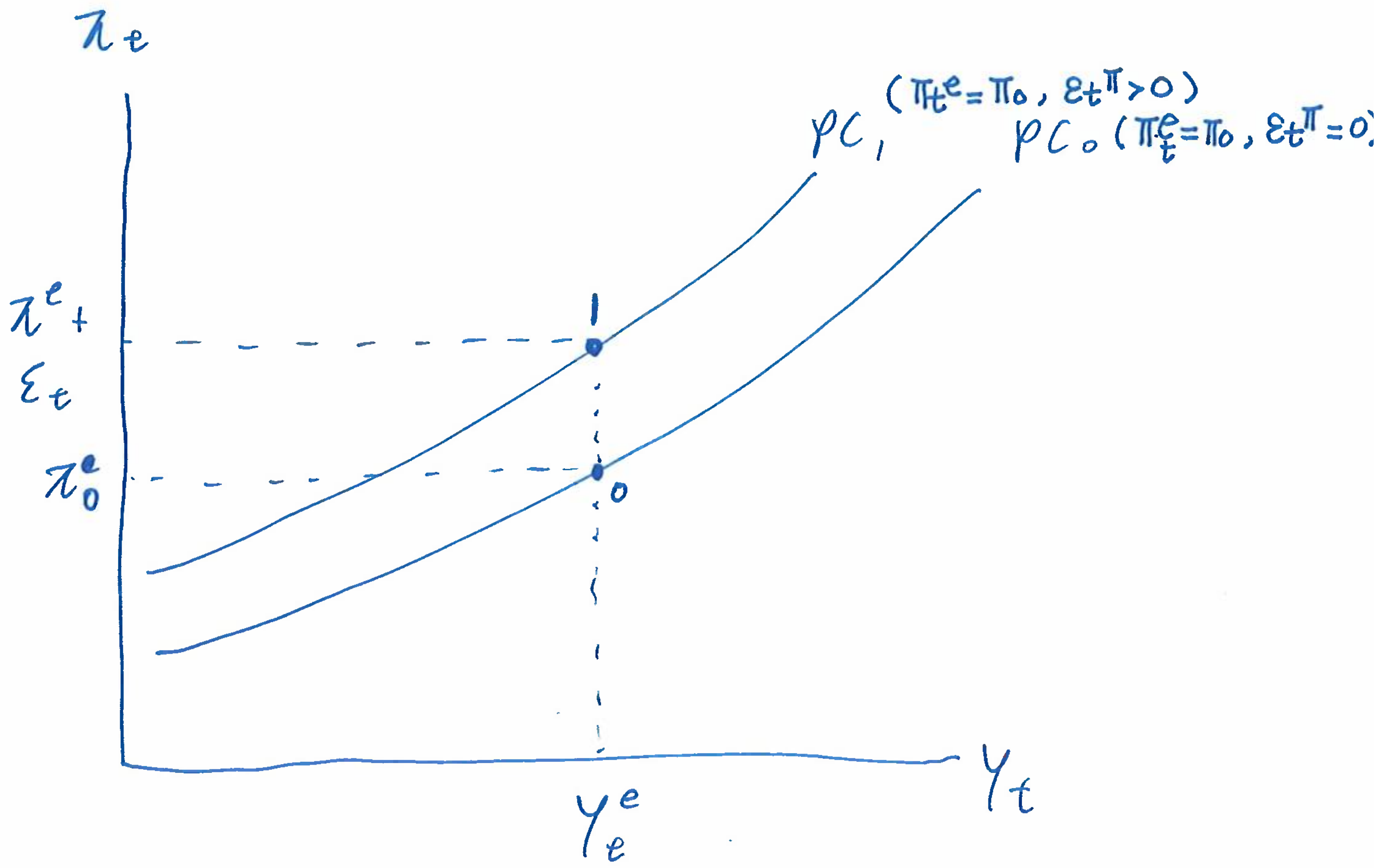
③ when $y_t < y_t^*$, $\pi_t < \pi_t^e$

y_t & π_t — endo

Shift in PC : π_t^e, y_t^*, ε_t^π exogenous variables

9. Figure 5 illustrates what happens when there is a positive inflationary shock so that ϵ_t^π goes from being zero to being positive. In this case, even when output equals its natural level ($y_t = y_t^*$) we still get inflation being above its expected level.

[**Figure 5:** The Phillips curve with $\epsilon_t^\pi > 0$]



10. Figure 6 illustrates how the curve changes when expected inflation rises from π_0^e to π_1^e. The whole curve shifts upwards because of the increase in expected inflation.

[**Figure 6:** The Phillips curve with higher π_t^e]